

Nightmares Course: V Tach

Section 1: Case Summary

Scenario Title:	Ventricular Tachycardia
Keywords:	Cardiology, ACLS, VT, Shockable Rhythm
Brief Description of Case:	In this scenario, the learner is called to the ward to assess a 65 y/o male with new VT. The learner must recognize the rhythm and institute appropriate work-up and management including electrical cardioversion.

Goals and Objectives	
Educational Goal:	Demonstrate an organized approach to the unstable patient.
Objectives: (Medical and CRM)	1. Recognize VT on an ECG 2. Initiate appropriate ACLS management for VT 3. Determine a safe disposition and transfer of care
EPAs Assessed:	

Learners, Setting and Personnel	
Target Learners:	☒ Junior Learners ☐ Senior Learners ☐ Staff
	☐ Physicians ☐ Nurses ☐ RTs ☐ Inter-professional
	☐ Other Learners:
Location:	☒ Sim Lab ☐ In Situ ☐ Other:
Recommended Number of Facilitators	Instructors: 1
	Confederates: 1 (RN)
	Sim Techs: 1

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Tim Chaplin
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Revised By:	Dr. Jared Baylis
Version Number:	1



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Surrinder Singh		Age: 65		Gender: M	
Weight: 90kg					
Presenting complaint: Palpitations					
Temp: 36.5	HR: 150	BP: 122/80	RR: 20	O ₂ Sat: 92%	FiO ₂ : room air
Cap glucose: 8.7			GCS: 15		
Triage note: You are called by the ward nurse to assess a 65 y/o male with a new onset "rapid heart rate". This patient was admitted early yesterday and is awaiting a coronary angiogram for an NSTEMI.					
Allergies: NKDA					
Past Medical History: Type 2 diabetes mellitus			Current Medications: ASA 81mg PO daily Metformin 1000mg PO BID		

Section 2B: Extra Patient Information

A. Further History	
There are no symptoms aside from a sensation of anxiety and apprehensiveness.	
B. Physical Exam	
List any pertinent positive and negative findings	
Cardio: Normal heart sounds, pulse rapid but palpable	Neuro: no neurologic deficits
Resp: nil	Head & Neck: nil
Abdo: nil	MSK/skin: nil
Other: The patient appears anxious.	



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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin – Adult
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Defibrillator BVM Non rebreather Nasal prongs Basic airway equipment
C. Required Medications
ACLS meds
D. Moulage
One PIV in the right antecubital fossa
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed
☐ Patient not yet on monitor
F. Patient Reactions and Exam
Patient is anxious and apprehensive Will endorse that he feels his heart racing Will deny any chest pain or shortness of breath

Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Ward Nurse	The ward nurse present in the room should have a headset to allow communication with the control room. They will be the nurse that called the resident to assess the patient and will be familiar with their past medical history and medications. Nurse will be helpful and non-obstructive.

Simulation Scenario Template

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: VT HR: 150 BP: 122/80 RR: 20 O ₂ SAT: 92% T: 37°C GCS: 15	Patient is anxious and apprehensive.	<u>Expected Learner Actions</u> ☐ Apply defibrillator pads ☐ Complete pulse check ☐ Order stat ECG and labs ☐ Begin ACLS algorithm of stable wide-complex tachycardia ☐ Apply supplemental oxygen ☐ Call for help ☐ Ensure IV access		<u>Modifiers</u> - Start amio/procaine/sotalol → no change to rhythm - Electrical cardioversion attempted → no change in rhythm - O ₂ → sat increase to 94% <u>Triggers</u> - 3 minutes or attempt at cardioversion → Phase 2
2. Unstable Rhythm: VT HR: 150 BP: 85/65 RR: 26 O ₂ SAT: 88% GCS: 14	Patient starts to complain of SOB and chest heaviness	<u>Expected Learner Actions</u> ☐ Recognize change in status ☐ Initiate ACLS for unstable tachycardia with a pulse ☐ Provide analgesia for shock ☐ Synchronized cardioversion ☐ Call cardiology/ICU		<u>Modifiers</u> <u>Triggers</u> - No shock within 3 minutes → Phase 3 - Shock given → Phase 4
3. Pulseless Rhythm: Pulseless VT HR: 150 (no pulse) BP: 0 RR: 0 O ₂ SAT: 0 GCS: 3		<u>Expected Learner Actions</u> ☐ Provide high quality CPR ☐ Provide BVM ventilation ☐ Consider intubation ☐ Provide defibrillation ☐ Call cardiology/ICU		<u>Modifiers</u> - - - <u>Triggers</u> - Defibrillation → Phase 4
4. Disposition Rhythm: Sinus HR: 90 BP: 110/80 RR: 18 O ₂ SAT: 94% on NP GCS: 15	Patient drowsy but GCS 15. Asking what happened.	<u>Expected Learner Actions</u> ☐ Recheck vitals and complete reassessment of patient post cardioversion ☐ Handover care to cardiology/ICU		

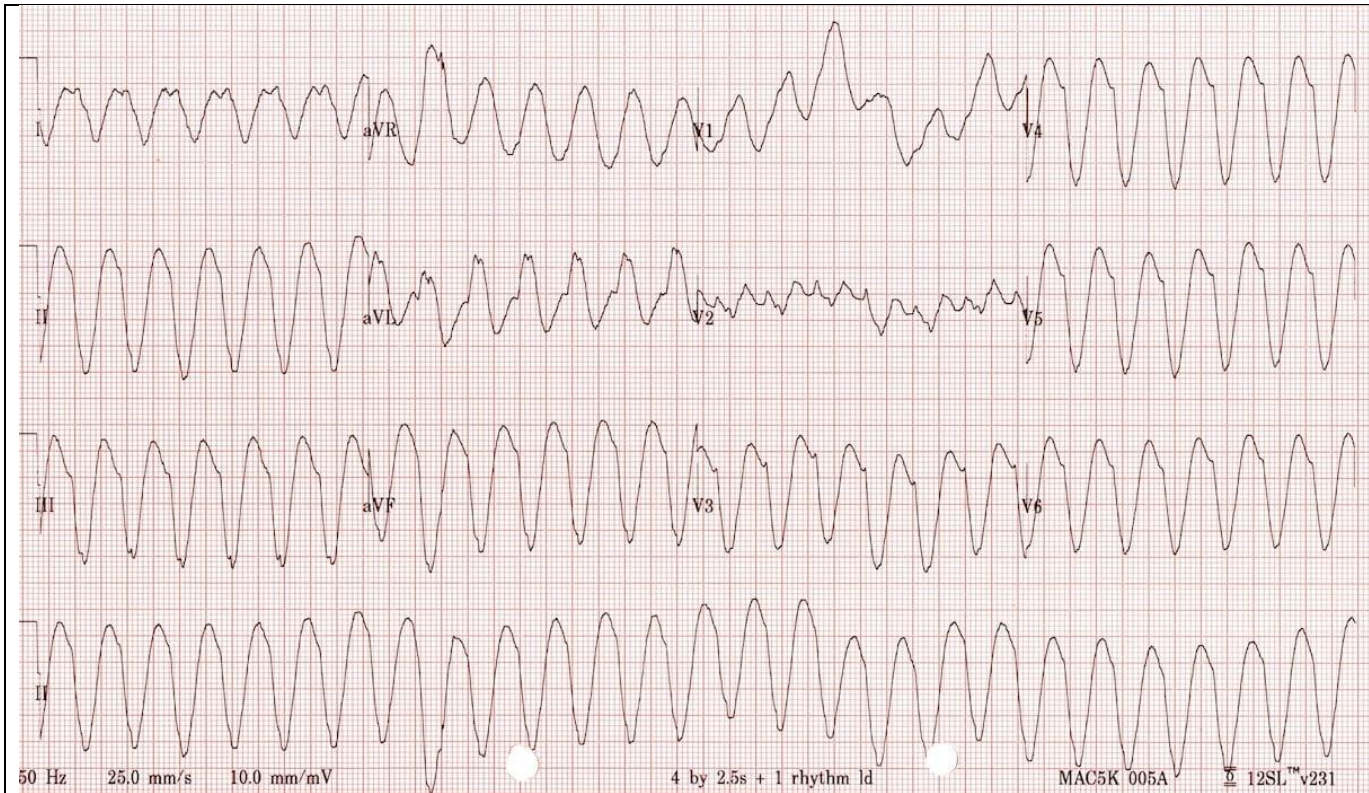


Simulation Scenario Template

Appendix A: Laboratory Results

*Note: No labs will be provided during this scenario.

Appendix B: ECGs, X-rays, Ultrasounds and Pictures



<https://litfl.com/ventricular-tachycardia-monomorphic-ecg-library/>

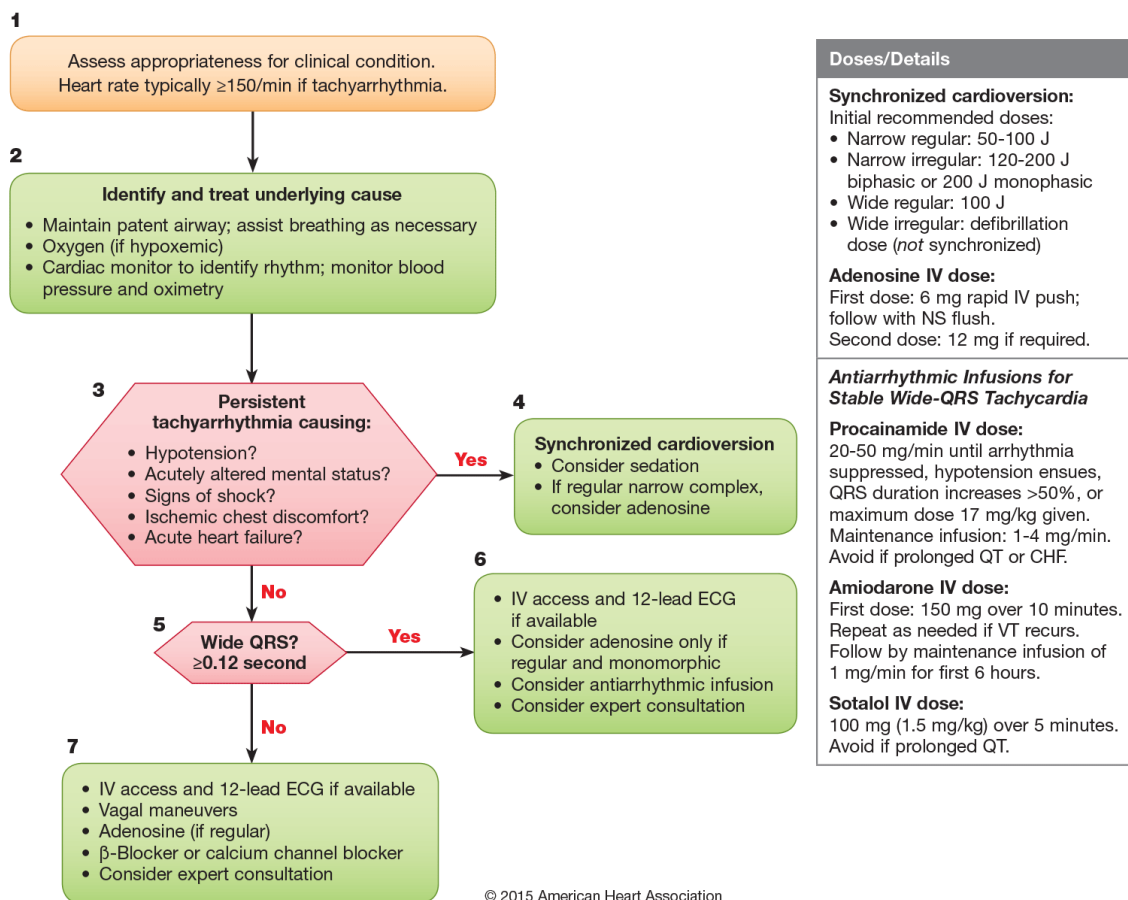
Simulation Scenario Template

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Debrief topics:

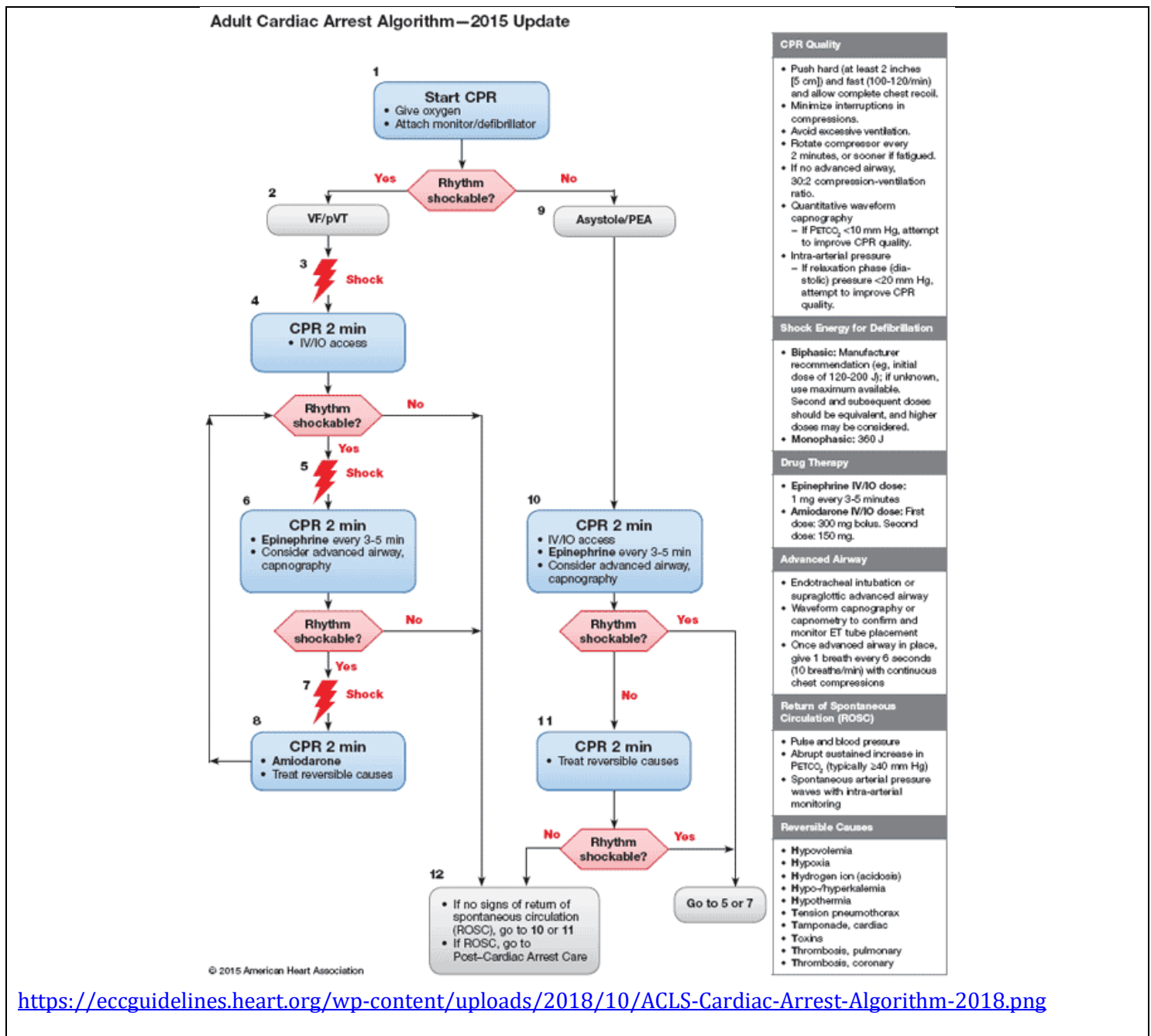
1. Clear communication within the team (RN, RT)
2. Initial approach to the stable patient
3. Criteria for the unstable patient
4. General approach to VT
5. Review of ACLS algorithms and supporting evidence

Adult Tachycardia With a Pulse Algorithm



https://eccguidelines.heart.org/wp-content/uploads/2015/10/2010-Integrated_Updated-Circulation-ACLS-Tachycardia-Algorithm.png

Simulation Scenario Template



References

1. ACLS 2015 Guidelines: <https://ahajournals.org/doi/10.1161/CIR.000000000000261>
2. ACLS 2019 Focused Update: <https://www.ahajournals.org/doi/10.1161/CIR.0000000000000732>
3. LITFL VTach: <https://litfl.com/ventricular-tachycardia-monomorphic-ecg-library/>

